MILITARY SPECIFICATION

PROJECTILE 30MM, HIGH EXPLOSIVE DUAL PURPOSE, M789, METAL PARTS

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts, equipment, requirements, quality assurance provisions and packaging for the 30MM High Explosive Dual Purpose Projectile for use in the 30MM, HEDP, M789 Cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (See 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Military

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

11825954	-	Cartridge, 30MM, HEDP, M789
9349963	-	Projectile, 30MM HEDP, M789, Metal Parts
11825929	-	Projectile, 30MM
11825958	-	Body, Projectile, 30MM, HEDP, M789
11825957	-	Body and Rotating Band Assy, Proj. 30MM, HEDP, M789

PUBLICATIONS

AMCR-715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedure for 20MM Cartridges

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the data of invitation for bids or request for proposal shall apply.

STANDARDS

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E18	-	Rockwell Hardness and Rockwell Superficial Hardness for Metallic Materials
ASTM B117	-	Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.2 Projectile. The projectile metal parts shall comply with all requirements specified on Drawing (Dwg.) 9349963, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.2 Manufacturing process. All parts and assembies shall be manufactured by a process approved by the contracting officer and no deviations from that process shall be made without prior approval. (see 6.7)

3.3 First article sample. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Banding. The rotating band blank and the rotating band seat shall be clean and dry at the time of banding. Shot or sandblasting shall not be used to clean the band seat after knurling. The rotating band blank shall fill the band seat.

3.5 Soundness. The projectile bodies shall be examined for metal discontinuities by eddy current or ultrasonic inspection. A discontinuity in excess of 0.25MM deep by 2.54MM in length shall be classified as a defect.

3.6 Protective coating.

3.6.1 Phosphate coating. The pretreatment process used shall be in accordance with applicable requirements of Dwg. 9349963. The phosphate coating shall be in accordance with the requirements of TT-C-490. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.6.2 Paint coating. The paint coating shall be in accordance with applicable requirements of Dwg. 9349963. Exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490 (see 6.8).

3.6.3 Corrosion resistance. The exterior painted surfaces of the projectile metal parts shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

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3.7 Metal parts security. The projectile metal parts, when assembled into test cartridges loaded in accordance with the requirements of Dwg. 11825954, shall not separate or break-up when tested for metal parts security.

3.8 Hardness. The projectile bodies shall comply with mechanical properties specified on the applicable drawings.

3.9 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.9.1 Process defects. The projectile metal parts shall be free of defects such as dents, deep scratches, cuts, sharp edges, burrs, metal flaws, excessive surface roughness and other forms of substandard workmanship.

3.9.2 Cleanliness. The projectile metal parts shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection (Table I)

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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PART DESCRIPTION	DRAWINGS	QUANTITY
Test panels (TT-C-490)		3
Blank rotating band for projectile 30MM	11825929	32
Body, Projectile - 30MM	11825958	32
Body and band assembly, HEDP	11825957	125
Projectile 30MM, HEDP, Metal Parts	9349963	400

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I.

4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings and specification requirements. The contractor's inspection shall be subjected to verification by the Government representative.

4.3.3 Rejection. See MIL-A-48078 and Table I herein.

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirements of the drawings and specifictions shall result in sample disapproval.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS DOD-P-63981 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Projectile Metal Parts		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Band (Dwg. 11825929)				
	Examination for defects	32		3.1	4.4.2.1
	Body, Projectile (Dwg. 11825958)				
	Examination for defects	32		3.1	4.4.2.2
	Body and Rotating Band Assy. (Dwg. 11825957)				
	Examination for defects	32		3.1	4.4.2.3
	Band tightness	125	0-1	3.4	4.5.1/4.4.3.1
	Projectile, Metal Parts (Dwg. 9349963)				
	Examination for defects	32**		3.1	4.4.2.4
	Soundness	125**	0-1	3.5	4.5.2/4.5.3/4.4.3.2
	Paint adhesion (see 6.8)	3**	0-1	3.6.2	4.5.5
	Corrosion resistance	32	0-1	3.6.3	4.5.6
	Metal parts security (firing test)	400	0-1	3.7	4.5.7/4.5.8/4.4.3.3
	Test panels per TT-C-490				
	Phosphate coating	3	0-1	3.6.1	4.5.4/4.4.3.4

NOTES:
** After examination these assemblies shall be used for band tightness test.
*** After examination these asemblies will be used for metal parts security.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, unless otherwise specified.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63981 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Blank Rotating Band for Projectiles 30MM	1 OF 1	11825929
			NEXT HIGHER ASSEMBLY 11825957

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major		See 4.4.2			
101	Defective metal, including lamination		.25%	3.1.1	Visual
102	Incomplete or deformed		.25%	3.1.1	Visual
Minor					
201	Width, maximum		.40%	3.1.1	Gage
202	Evidence of poor workmanship		.65%	3.4/3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63981 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Body, Projectile, HEDP		11825958
			NEXT HIGHER ASSEMBLY 11825957

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major		See 4.4.2			
101	Length, mouth to rear of rotating band seat		.25%	3.1	Gage
102	Knurl, missing or incomplete		.25%	3.1	Visual
103	Foreign matter in rotating band seat		.25%	3.1/3.4	Visual
104	Diameter, rotating band seat		.25%	3.1	Gage
105	Bourrelet diameter base, min		.25%	3.1	Gage
106	Crimp groove profile		.25%	3.1	Comparator
107	Cavity diameter max		.25%	3.1	Gage
108	Runout, cavity diameter with datum surface		.25%	3.1	Gage
109	Length, base to rear of band seat		.25%	3.1	Comparator
110	Minor diameter thread		.25%	3.1	Gage
111	Runout, thread with datum surface		.25%	3.1	Gage
112	Thread depth (length), max.		.25%	3.1	Gage
113	Base chamfer		.25%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63981 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Body, Projectile, HEDP		11825958
			NEXT HIGHER ASSEMBLY 11825957

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Width, rotating band seat		.40%	3.1	Gage
202	Profile mouth diameter and mouth radius		.40%	3.1	Comparator
203	Finish improper		.40%	3.1	Visual
204	Length, complete threads, min		.40%	3.1	Gage
205	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63981 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Body and Rotating Band Assembly, HEDP	1	1	11825957
				NEXT HIGHER ASSEMBLY 9349963

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Soundness		100%	3.5	4.5.2/4.5.3/4.4.3.2
Major		See 4.4.2			
101	Length, mouth to rear edge of rotating band		.25%	3.1	Gage
102	Diameter bourrelet, min		.25%	3.1	Gage
103	Diameter rotating band		.25%	3.1	Gage
104	Runout, rotating band w/datum surface		.25%	3.1	Gage
105	Gap, junction rotating band and body		.25%	3.4	4.4.3.6
106	Band tightness	20	ACC-0 REJ-1	3.4	4.5.1 (See Note)
Minor		See 4.4.2			
201	Depth and width of undercut, front and rear of rotating band, max		.40%	3.1	Profile comparator chart (minor 201-202)
202	Width, rotating band		.40%	3.1	
203	Corrosion		.40%	3.1	Visual
207	Evidence of poor workmanship		.65%	3.9	Visual

NOTES: If crimp groove configuration is machined at this operation refer to para. 4.4.2.2 CD 108 & 110.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63981 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Projectile, 30MM, HEDP, M789, Metal Parts				9349963
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See 4.4.2a			
101	Foreign matter in cavity or threads		.25%	3.1	Visual
102	Phosphate coating (panels)	3	ACC-0 REJ-1	3.6.1	4.5.4/4.4.3.4
103	Paint adhesion	2	ACC-0 REJ-1	3.6.2	4.5.5
104	Metal parts security (firing test)	75	ACC-0 REJ-1	3.7	4.5.7/4.5.8/4.4.3.3

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Projectile, 30MM, HEDP, M789, Metal Parts				9349963
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor		See 4.4.2a			
201	Interior or exterior coating inadequate		.40%	3.1	Visual
202	Paint on threads		.40%	3.1	Visual
203	Rotating band damaged		.40%	3.1	Visual
204	Foreign matter on exterior		.40%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.9	Visual
206	Diameter rotating band, max		.40%	3.1	Gage
207	Diameter bourrelet, max		.40%	3.1	Gage
208	Diameter base, max		.40%	3.1	Gage

NOTES:

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4.4.3 Testing.

4.4.3.1 Sand fill. Projectile assemblies shall be sectioned through the longitudinal axis to expose the band fill in the undercut areas. Each undercut area so exposed shall as a minimum be 50% filled by the rotating band material.

4.4.3.2 Inspection verification. Inspection verification by soundness inspection shall be performed by utilization of Government approved machine standards. Rejection of one or more units of the sample by the flow detector in the test for body metal soundness shall be cause for rejection of the lot. MIL-STD-105 shall be used to determine sample size (minimun sample size = 315), except for First Article Testing.

4.4.3.3 Metal parts security. The occurrence of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of lot, except as otherwise provided:

a. Body breakup.

b. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

c. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

d. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less assemblies throw band fragments having a length not greater than 13mm.

(2) The lot will be rejected if any assembly throws a band fragment having a length greater than 13mm (indicative of deep-seated band lamination).

(3) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.3.4 Phosphate coating. Failure of the phosphate coating on the standard panels or sample body assemblies to comply with the applicable requirements shall be cause for rejection of all projectiles phosphated since the preceding test.

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4.4.3.5 Rotating band blank. A rotating band blank which is deformed or judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions shall be classed defective.

4.4.3.6 Gap. A .254MM feeler gage shall not enter the gap between rotating band and projectile body, both forward and aft 360°.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Band tightness. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas. Failure of either of the exposed sections to comply with the 50% fill requirements shall be considered a defect.

4.5.2 Soundness. The procedures for conducting the soundness inspection shall be prepared by the contractor and submitted to the contracting officer for approval. The procedure shall make provision for the use of defect standards.

4.5.2.1 Defect standards. As a minimum, the defect standards used in the soundness test will be cycled through the soundness test equipment every shift. Failure of the test equipment to reject the defect standards shall be cause for rejection of assemblies inspected for soundness since the preceding test of the defect standards.

4.5.3 Automatic inspection equipment calibration verification. The defect standards for soundness shall be cycled through the soundness inspection equipment at the start and completion of each shift as well as after any adjustment or shutdown of the soundness inspection equipment. Failure of the equipment to reject standards shall be cause for rejection of assemblies tested for soundness since the preceding satisfactory test for defect standards.

4.5.4 Phosphate coating. The method of test shall be as prescribed by TT-C-490 for the type of coating used (see Dwg. 9349963). The prescribed panels shall be processed, with the projectiles to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

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4.5.5 Paint adhesion. The method of test shall be prescribed by TT-C-490. The test shall be made on the painted bourrelet of each unit of the sample, forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal of any unit of the sample, as a result of the test, shall be recorded as a defect.

4.5.6 Corrosion resistance. The method of test shall be prescribed in American Society of Testing and Materials (ASTM), ASTM B117. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.7 Metal parts security. The method of test shall be as prescribed in AMCR-715-505, Volume 8, as applicable. The target shall be 8' by 8' 1" thick plywood with 1/2" thick celotex facing. Target shall be located 5 meters ± 1 meter from the muzzle. The test sample shall be assembled to comply with requirements of Dwg. 11823954, Cartridge, 30MM, for projectile metal parts test. The test cartridges shall be fired at ambient temperature in bursts of 25 shots. The gun barrel may be cooled to ambient temperature after each 100 rounds. The cyclic rate of each burst shall be the design cyclic rate.

4.5.7.1 Defective weapons. If any sample projectile firing defect is found to have been caused by a defective test weapon, the firing defect shall not be classed as a failure. In such case, the defective test weapons shall be corrected or replaced and the test repeated in whole or in part, as appropriate. If such firing defect is not found to have been caused by a defective weapon, it shall be classed as a failure.

4.5.8 Test gun. Unless otherwise directed by the Contracting Officer, the metal parts security firing tests shall be conducted in the 30MM, XM230 Gun.

5. PRESERVATION, PACKAGING AND PACKING

5.1 Packaging and packing, level C. Each lot of projectile metal parts shall be packed using best commercial packaging and packing as approved by the procuring agency.

5.2 Marking and labeling. Marking or labeling of packed cartons or containers shall comply with applicable requirements of MIL-STD-1169.

5.3 Packaging, packing and marking inspection. Inspection for packaging, packing and container marking shall be in accordance with the quality assurance provisions of MIL-STD-1169.

5.4 Shipping. When shipments of more than one lot are made at one time, such as in a carload, each lot shall be kept separate and the division between lots clearly indicted to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended use. This specification covers projectile, metal parts and packing for Projectile, 30MM, HEDP, M789.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment design as required to: Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. Request letter of submittal, state contractor, contractor number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor-conducted examinations and test. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commmander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridges are procured by ARRADCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwrded to: Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

6.6 First article. When a first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, test and approval of the documents for first article.

6.7 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture or an operational change which alters the metallurgical or physical properties of the item of contract.

6.8 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pre-treatment shall be considered unsatisfactory if any of the following conditions exist.

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters.

c. More than five areas of 15 square millimeters or less.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A889)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-P-63981 (AR) PROJECTILE 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789, METAL PARTS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
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